Relativistic Corrections in Satellite Positioning Systems

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Contents

1. Introduction

Satellite positioning systems, or GNSS (Global Navigation Satellite Systems), are designed to determine the position of any observer on the Earth's surface and are based on a positioning principle known as trilateration, which consists of measuring different distances to points whose locations are already known.

If a constellation of satellites in orbit emits signals at a given speed, any observer on the ground can determine the distance separating them from each satellite by measuring the time it takes for the signal to arrive. Thus, through the trilateration resulting from several distances, the observer can determine the specific point at which they are located with respect to the satellites, and by knowing the satellites' positions, they can also determine their own position on the surface.

To achieve this, a measurement of that time is required, and in the case of perfect time synchronization, it would be enough to send a single synchronous timestamp with all signals so that the observer could compare the received times with their own and obtain those distances.

However, satellites move at high velocities and altitudes above the Earth's surface, making relativistic effects significant, and these effects are further amplified by the propagation speed of the signal, which is the speed of light in vacuum.

Under these conditions, if no correction were introduced for the measurement of proper time on board the satellites, synchronization would quickly be lost after any resynchronization event, making it impossible to establish those distances reliably and to determine position accurately from them.

It is important to highlight that the relativistic correction is not the only one required in satellite geolocation, because there are other effects that introduce measurement errors on the ground—for example, those derived from the rotation of the observer comoving with the Earth and affected by its rotation (Sagnac effect), or those caused by the chemical composition of the atmosphere, which delays the signals.

The purpose of this document is to detail the relativistic corrections required due to altitude and velocity. It also describes the basic principles of trilateration for satellite positioning and includes several demonstrations and notes on analytical mechanics that may be useful in the context of general relativity and satellite navigation.

All images included in this document are either created by the author or generated using Google's Nano Banana 2. For all numerical approximations and other underived expressions common to both special and general relativity, the following sources have been consulted

Sources	
#1	Spacetime and Geometry: An Introduction to General Relativity, Carroll, S. M.
#2	Relativity in the Global Positioning System, Ashby N.
#3	Relativistic Corrections in the European GNSS Galileo, ESA, Mudrak, A., De Simone, P., Lisi, M. (2015)
#4	Geodetic Reference System 1980, IUGG
#5	Solar System Dynamics, JPL
#6	GPS Space Segment (ICD), U.S. Department of Defense
#7	GLONASS Interface Control Document (ICD), UNAVCO
#8	Galileo Interface Control Document (ICD), European GNSS Service Centre
#9	BeiDou Interface Control Document (ICD), China Satellite Navigation Office (CSNO)

2. Trilateration

2.1. Trilateration in the Plane

Let us imagine a plane and a circle C_1 whose points are all at a distance r_1 from its center o_1. No observer on the circumference can determine their exact position by measuring the radius r_1—only that they lie somewhere on the circle. Now imagine another circle with radius r_2 and center o_2 at an arbitrary distance from o_1. Both circles have two intersection points, p_1 and p_2 (figure 1).

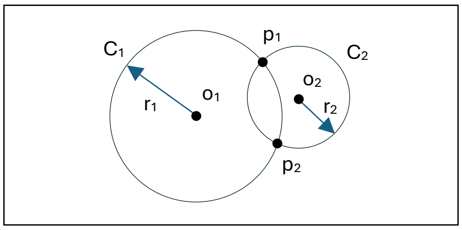


Figure 1

We can see that by knowing r_1 and r_2, the observer can determine that they are either at p_1 or p_2 with respect to o_1 and o_2, but not which one. We can also see from the very definition of the problem that there are infinitely many circles passing through those two points, as many as can be drawn with centers on the line defined by the centers of the two circles, since that line is the perpendicular bisector of the segment joining p_1 and p_2 (figure 2).

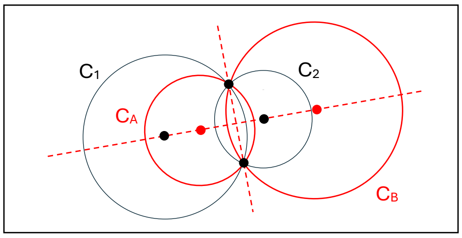


Figure 2

Now introduce a third circle C_3 provided its center does not lie on the line joining the previous two centers (figure 3).

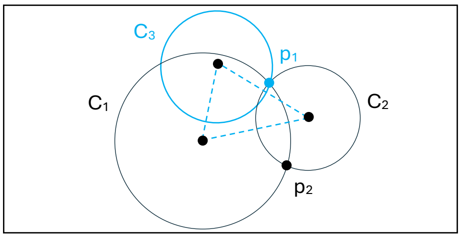


Figure 3

We see that there is no circle that, while passing through one of the intersection points of two circles, can also pass through the other: point p_1 is thus determined by the three radii of circles C_1, C_2, and C_3. Determining a position in the plane using this procedure from three distances to points of known position is called trilateration.

2.2. Trilateration in Space

Let us now try to extend this process to our three-dimensional space. Observe that the equivalent of a circle in the plane is the surface of a sphere S_1, where all points are at a distance r_1 from its center o_1. No observer on the surface of the sphere can determine their exact position—only that they lie somewhere on that surface. Now imagine another sphere S_2 with radius r_2 and center o_2 at an arbitrary distance from o_1. The surfaces of both spheres share an entire circle as their intersection, C_{12} (figure 4).

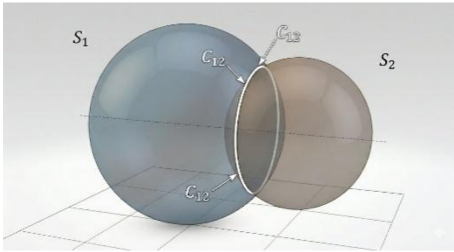


Figure 4

Again, from the definition of the problem, there are infinitely many spheres whose surfaces share the intersection C_{12}, provided their centers lie on the line joining the two original centers, since that line is the axis perpendicular to the plane of the intersection circle C_{12} passing through its center (figure 5).

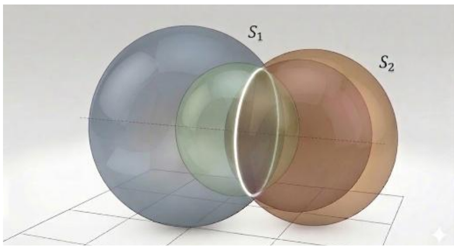


Figure 5

Now introduce a third sphere S_3 provided its center does not lie on the line joining the centers of the previous two. The intersection of the surfaces of the three spheres is now reduced to two points, which are the intersection of the surface of S_3 with the circle resulting from the intersection of the other two, C_{12} (figure 6).

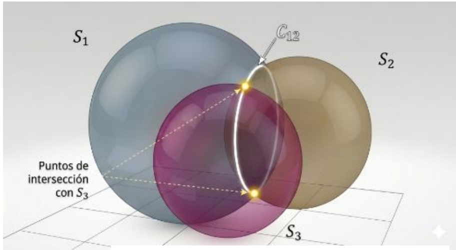


Figure 6

However, both points are equally possible, making a fourth sphere S_4 necessary. That fourth sphere cannot lie in the same plane as the centers of the other three, or we would again face a similar indeterminacy, finding infinitely many spheres whose surfaces intersect at those two points. We see that it is impossible to find a fourth sphere whose center does not lie in that plane and whose surface passes through both intersection points of the previous three surfaces, so point p_1 becomes fully determined (figure 7).

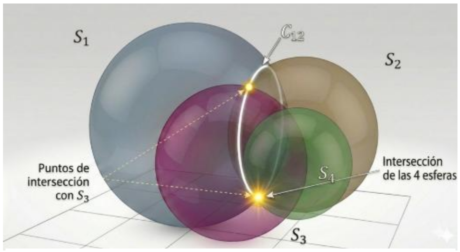


Figure 7

This is the positioning system in space used by GNSS systems.

2.3. Satellite Trilateration

Here it is worth noting that when this procedure is used to determine satellite-based geolocation, the Earth itself resembles a sphere, so it would be sufficient to use three additional spheres (three satellites) to determine the position. An observer on the Earth's surface would know which of the two possible intersection points of the other three sphere surfaces corresponds to their location, because the other intersection point would be inaccessible, thus determining their position (figure 8).

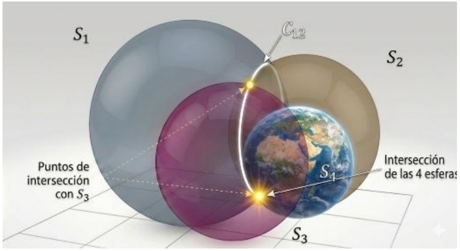


Figure 8

However, at least four satellites are used for reasons related to the ability to measure the observer's proper time on the ground, which lacks the required precision and synchronization.

We know how to trilaterate a position in space using four fixed points, but how do we solve trilateration when the centers of the spheres are moving? This is the case for satellites, whose velocities are significant and which are no longer at the same position relative to the receiver on the surface by the time their signals are received.

Thus, satellites must transmit not only a timestamp but also their position at the moment of emission. Once other corrections are applied, the geolocation problem becomes one of determining the position of points on the surface relative to the satellites.

Satellites therefore act as position references, self-determined from the stability of their orbits, and broadcast in the form of ephemerides, i.e., orbital parameters describing their trajectory at each moment. They also require corrections for any perturbations in their orbital mechanics. To ensure the accuracy of all measurements—both position and time—periodic updates are sent to the satellites from the ground.

For this reason, medium Earth orbits (MEO), non-geostationary and with periods shorter than one day, are used. These allow frequent updates from the ground while keeping satellites in free fall, as will be detailed later. As a point of interest, satellites also broadcast the constellation's almanac, i.e., not only their own ephemerides but also those of the other satellites, helping the ground receiver identify and validate signals from other satellites at any given moment.

3. Relativistic Corrections

We now proceed to specify the two relativistic corrections that must be implemented at the level of the satellites' proper-time clocks. Among all required corrections, these two are the only ones that are common to all observers on Earth's surface, regardless of their location. For this reason, they are the only corrections configured directly into the satellites' clock offsets before launch.

The remaining corrections do not apply uniformly across the surface and are therefore handled in software at the receiver level. For example, the correction required due to Earth's rotation (Sagnac effect) depends on the observer's latitude: surface velocity is maximal at the equator and zero at the poles. Similarly, atmospheric delays depend on the chemical composition of the atmosphere, which is not homogeneous across the globe.

Furthermore, among all necessary corrections, the non-relativistic ones (Sagnac, atmospheric composition, etc.) represent only a small fraction of the total effect—although they cannot be neglected if precise satellite navigation is desired. The two relativistic effects account for almost the entire correction, with the gravitational term being the dominant contribution.

3.1. Imprecision in the Observer's Time Measurement on the Ground

All signals travel at the speed of light in vacuum ($\sim 10^8$ m/s), meaning that microscopic timing errors translate into unacceptable distance errors. If we compute the distance to satellites using

(3.1) $d = c \cdot \Delta t$

we see that each 1 μs of error ($\sim 10^{-6}$ s) results in a distance error of hundreds of meters ($\sim 10^2$ m), and this is for a single satellite. Such errors would accumulate with relativistic effects until corrected again from the ground, severely limiting the system's precision and usefulness.

For this reason, satellites are equipped with highly precise proper-time measurement technologies such as atomic clocks. However, ground receivers cannot be equipped with such technology. This is why we do not use only three satellites to determine surface position, but four.

Using a fourth satellite allows the receiver to compute position accurately without depending on the observer's own imperfect time measurement. Moreover, it enables portable devices—without precise clocks—to benefit from the timing precision of the satellites' atomic clocks. In practice, even more satellites are used to mitigate possible defects or reception errors (e.g., multipath reflections in urban environments).

Ground observers compute distances to satellites using the equations:

(3.2) $(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = {d_i}^2 = (c \cdot \Delta t_i)^2 = c^2 \Delta t_i^2$

where x_i is the position of satellite i, d_i the distance, and x the observer's position. Let b be the difference between the observer's proper time and the satellites' time (assuming perfect synchronization among satellites). Then:

(3.3) $(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = {d_i}^2 = c^2 \cdot (t_i - \mathrm{t})^2 = c^2 b^2$

and therefore:

(3.4) $\sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} - \mathrm{c} \cdot b = 0$

We can write one such equation for each satellite. The sign of the time difference does not matter, since it is absorbed by the squared distance.

Thus, we have a system of four equations with four unknowns, one of which (the clock bias b) is common to all. Four satellites allow us not only to determine the position but also to solve for b, enabling the receiver to adjust its own time and effectively inherit the precision of the satellites' atomic clocks.

3.2. Relativistic Corrections Due to Velocity

We know that the proper time of two observers in motion tick at different rates depending on their relative velocities. Ideally, we would compute the velocities of both satellites and ground observers in the same reference frame. However, surface velocity is not uniform, so we adopt a reference frame centered at Earth's center that does not rotate with it: the Earth-Centered Inertial (ECI) frame.

In this frame, ground observers are nearly static compared to satellites, and software corrections are applied at the receiver level. As we will see numerically, the ratio between orbital velocities and surface velocities is indeed large, validating this approach for special-relativistic treatment in satellite navigation.

3.2.1. Free-Fall Velocity (Perfectly Circular Orbit)

For Earth's gravitational field, we use the Schwarzschild metric, treating Earth as a perfect radial symmetry:

(3.5)

$$ds^2 = -c^2\left(1-\frac{r_s}{r}\right)dt^2 + \left(1-\frac{r_s}{r}\right)^{-1}dr^2 + r^2 d\Omega^2 \quad \text{where } r_s = \frac{2GM}{c^2}$$

This solution imposes an asymptotically flat spacetime, which is already weak under satellite operating conditions, where Newtonian mechanics would suffice.

By spherical symmetry, we restrict motion to the equatorial plane ($\theta = \pi/2$, $\dot{\theta} = 0$). The spatial velocity components in this plane are the radial velocity v_r and the tangential velocity v_l (associated with the angular coordinate ϕ). Without symmetry, two angular components would appear.

This is consistent with the Christoffel symbols under these conditions:

(3.6)

$$\Gamma^{\phi}_{\theta\theta} = 0$$

$$\Gamma^{\phi}_{\theta\phi} = \Gamma^{\phi}_{\phi\theta} = \cot\theta = \cot\pi/2 = 0$$

Using the radial geodesic equation:

(3.7)

$$\ddot{r} - \frac{r_s}{2r(r-r_s)}\dot{r}^2 - r\left(1-\frac{r_s}{r}\right)\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + c^2\frac{r_s}{2r(r-r_s)}\dot{t}^2 = 0$$

and evaluating at the equatorial plane

(3.8)

$$\ddot{r} - \frac{r_s}{2r(r-r_s)}\dot{r}^2 - r\left(1-\frac{r_s}{r}\right)\dot{\phi}^2 + c^2\frac{r_s}{2r(r-r_s)}\dot{t}^2 = 0$$

we obtain the tangential velocity by imposing $\dot{r} = 0$ and dividing by $\dot{t}^2$:

(3.9)

$$-r\left(1-\frac{r_s}{r}\right)\dot{\phi}^2 + c^2\frac{r_s}{2r(r-r_s)}\dot{t}^2 = 0$$

$$\Rightarrow -r\left(1-\frac{r_s}{r}\right)\frac{v_l^2}{r^2} + c^2\frac{r_s}{2r(r-r_s)} = 0$$

which yields:

(3.10)

$$v_l^2 = \frac{GMr}{(r-r_s)^2} \Rightarrow v_l = \frac{\sqrt{GMr}}{r-r_s}$$

which is the coordinate tangential velocity in circular orbits. In the calculation of time dilation we must use the local velocity but in the case of satellite operation we find that this expression is valid because they orbit far from r_s ($r \gg r_s$) so we can consider that local velocity equals the coordinate velocity. We can further simplify the above expression and therefore use $(r - r_s) \approx r$

(3.11)

$$v_l = \frac{\sqrt{GMr}}{r} = \sqrt{\frac{GM}{r}}$$

This matches the Newtonian orbital velocity obtained by equating centripetal and gravitational forces:

(3.12)

$$F_g = F_c \Rightarrow G\frac{Mm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

3.2.2. Relativistic Correction in Free Fall (Perfectly Circular Orbit)

We now compute the time dilation experienced by static observers relative to moving ones, using the Lorentz factor:

(3.13)

$$\gamma = \frac{dt}{d\tau} = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

which for $v \ll c$ becomes

(3.14)

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} \approx \frac{1}{1-\frac{1}{2}\frac{v^2}{c^2}}$$

Thus, the difference in proper-time flow is

(3.15)

$$\Delta\tau_{\text{velocity}} = \frac{dt - d\tau}{dt} = 1 - \frac{d\tau}{dt} = 1 - \frac{1}{\gamma}$$
$$= 1 - \left(1 - \frac{1}{2}\frac{v^2}{c^2}\right) = \frac{1}{2}\frac{v^2}{c^2}$$
$$= \frac{GM}{2rc^2}$$

This has a positive sign but represents a delay in the proper time of moving observers relative to static ones.

3.2.3. Velocity Correction Applied to Satellite Geolocation

To obtain the difference in time dilation between satellites and ground observers, we should use their velocity difference. However, due to the non-uniformity of surface velocities, we use only the orbital velocity relative to the ECI frame.

The ratio between satellite velocity v_{sat} and surface velocity v_{Earth} is:

(3.16)

$$\frac{v_{\text{sat}}}{v_{\text{Earth}}} = \frac{\omega_{sat}}{\omega_{\text{Earth}}}\frac{(r_t + h)}{r_t} = \frac{\omega_{sat}}{\omega_{\text{Earth}}}\left(1 + \frac{h}{r_t}\right)$$

where h is satellite altitude.

Since satellites have an orbital period of roughly half a day, $\omega_{sat} \approx 2 \times \omega_{\text{Earth}}$. Also, for GNSS, $\mathrm{h} \gtrsim 3 \times \mathrm{r_t}$. Thus:

(3.17)

$$\frac{v_{sat}}{v_{\text{Earth}}} \gtrsim 2 \times (1+3) = 8$$

Therefore:

(3.18)

$$\Delta\tau_{\text{velocity}} = \frac{1}{2c^2}\left({v_{sat}}^2 - {v_{\text{Earth}}}^2\right) \approx \frac{1}{2c^2}\left({v_{sat}}^2 - \frac{{v_{sat}}^2}{64}\right)$$

where the second term is only ~1.5% of the first and can be neglected:

(3.19)

$$\Delta\tau_{\text{velocity}} = \Delta\tau_{vel\ \text{sat}} - \Delta\tau_{vel\ \text{Earth}} \approx \Delta\tau_{vel\ \text{sat}} = \frac{1}{2c^2}{v_{\text{sat}}}^2 = \frac{GM}{2rc^2}$$

And using GRS80 values (#4):

(3.20)

$$\mathrm{v}_{\text{Earth}\ equator} = \omega_{\text{Earth}} \times r_{equator} = 7.292115 \times 10^{-5} \times 6\,378\,137 = 465.1\ \text{m/s}$$

Let us check a specific case. For a satellite at 24,385 km altitude:

(3.21)

$$\varepsilon(\%) = \frac{({v_{\text{sat}}}^2 - {v_{\text{Earth}}}^2)}{{v_{\text{sat}}}^2} \times 100 = \frac{(3\,600^{\,2} - 465.1^{\,2})}{3\,600^{\,2}} \times 100 = 1.67\%$$

Thus:

(3.22)

$$\Delta\tau_{\text{velocity}} = \frac{GM}{2(r_t + h)c^2}$$

or, using the Schwarzschild radius:

(3.23)

$$\Delta\tau_{\text{velocity}} = \frac{GM}{2(r_t + h)c^2} = \frac{r_s}{4(r_t + h)}$$

3.3. Relativistic Corrections Due to Gravity

We know that the proper time of two observers in a gravitational field tick at different rate depending on their difference in gravitational potential. We now want to compute these differences for Earth's gravitational field.

3.3.1. Relationship Between Proper Time and Gravitational Potential in Spherical Symmetry

We use the same working metric (Schwarzschild), expressed as:

(3.24)

$$ds^2 = -c^2\left(1 - \frac{r_s}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2 \quad \text{where } r_s = \frac{2GM}{c^2}$$

Spherical static symmetry implies that the potential does not depend on angular coordinates or on coordinate time—only on the radius. It is important to note that we are not trying to define the spacetime interval between two events at different radii, but rather the difference in proper-time flow relative to coordinate time at two different radii.

Since g_{tt} already relates proper time and coordinate time, setting $dr = d\Omega = 0$ yields:

(3.25)

$$d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2 \Rightarrow \frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

Because r and t are independent variables, r is constant during integration over time. Therefore:

(3.26)

$$\Delta\tau_{\text{gravity}} = \sqrt{1 - \frac{r_s}{r}} \Delta t$$

For weak fields, we approximate:

(3.27)

$$\Delta\tau_{\text{gravity}} = \sqrt{1 - \frac{r_s}{r}} \Delta t \approx \left(1 - \frac{r_s}{2r}\right) \Delta t$$

This expression allows us to compute the difference in proper-time flow between two radii r_a *and* r_b with $r_b > r_a$:

(3.28)

$$\begin{aligned}\Delta\tau_{\text{gravity}} = \Delta\tau_{grav\, r_b} - \Delta\tau_{grav\, r_a} \\ \approx \left(1 - \frac{r_s}{2r_b}\right) - \left(1 - \frac{r_s}{2r_a}\right) \\ = \frac{r_s}{2r_a} - \frac{r_s}{2r_b}\end{aligned}$$

This can be generalized for any radius r and increment Δr:

(3.29)

$$\Delta\tau_{\text{gravity}} = \frac{r_s}{2r} - \frac{r_s}{2(r + \Delta r)}$$

3.3.2. Gravitational Correction Applied to Satellite Geolocation

We can rewrite expression 3.29 using the absolute value of the classical Newtonian potential $V = -\frac{GM}{r}$, substituting the Schwarzschild radius and assigning the values for Earth's radius r_t and the satellite orbital radius r_{sat}:

(3.30)

$$\begin{aligned}\Delta\tau_{\text{gravity}} = \frac{r_s}{2r_t} - \frac{r_s}{2r_{sat}} = \frac{GM}{r_t c^2} - \frac{GM}{r_{sat} c^2} \\ = \frac{V_t}{c^2} - \frac{GM}{r_{sat} c^2}\end{aligned}$$

Here,

V_t is the absolute value of the gravitational potential at Earth's surface. This substitution is convenient because both this value and Earth's mean radius are explicitly included in the GRS80 mode (#4).

3.3.3. Isolating the Effects of Nearby Masses

It seems appropriate to check whether masses near the Earth influence the GNSS system by creating significant potential differences between the surface and the satellites beyond those created by the Earth's field. Fortunately, this is not the case, and we will verify that their influence is indeed negligible.

In the EFE in its general form:

(3.31)

$$R_{ab} - \frac{1}{2} R g_{ab} = G_{ab} = \frac{8\pi G}{c^4} T_{ab}$$

all coupled equations must be satisfied. For combined systems of n gravity sources, all equations of the following form would have to be satisfied:

(3.32)

$$G_{ab}^{(n)} = R_{ab}^{(n)} - \frac{1}{2} R^{(n)} g_{ab}^{(n)} = \frac{8\pi G}{c^4} T_{ab}^{(n)}$$

which, as we see, can never be generalized as a system of equations because the curvature tensors would belong to different metrics; thus, we cannot express the curvature as a linear expression of those generated by the different masses.

Verifying that we can discriminate the influence of other masses in the solar system thus becomes very difficult unless all fields are weak and we can sum potentials in a Newtonian manner. That is, by assuming the curved metric as the sum of a flat metric and a minimal perturbation that allows us to linearize it, in the form:

(3.33)

$g_{ab} = \eta_{ab} + x_{ab}$ with $x_{ab} \ll \eta_{ab}$

where η_{ab} is the flat Minkowski metric, g_{ab} represents the metric we want to linearize, and x_{ab} is considered the perturbation that adds curvature. In this way, the equations remain linearized in the weak-field regime.

The problem is simplified by confirming that all masses in the solar system generate weak fields and that their influence on Earth, which can now be summed, is negligible compared to the terrestrial field. Let us check these fields for each significant mass in the solar system.

The Sun's influence, by the equivalence principle, will only manifest in tidal forces. Given that these tidal forces manifest with a value inversely proportional to the cube of the distance and that the average of that distance is by definition 1 UA ≈ 150 $\times 10^9$ m (#5, glossary), we can assume them to be negligible.

For the representative planets and satellites of the solar system, ordering their influence from greatest to least on the Earth's gravitational field—with the Earth's potential at its surface being $\Phi_{Earth} = -6{,}26 \times 10^7$ J/kg (#4, Derived Physical Constants) and according to JPL values (#5, Horizons, Astrodynamic Parameters)—we can construct the following table (Table 1)

Body	Min D (km)	Φ at D (J/kg)	Φ/Φ_{Earth}
Jupiter	$5{,}88\times10^8$	$-2{,}15\times10^5$	$3{,}44\times10^{-3}$
Saturn	$1{,}20\times10^9$	$-3{,}16\times10^4$	$5{,}06\times10^{-4}$
Moon	363.300	$-1{,}35\times10^4$	$2{,}16\times10^{-4}$
Venus	$3{,}82\times10^7$	$-8{,}55\times10^3$	$1{,}35\times10^{-4}$
Uranus	$2{,}57\times10^9$	$-2{,}25\times10^3$	$3{,}60\times10^{-5}$
Neptune	$4{,}30\times10^9$	$-1{,}59\times10^3$	$2{,}54\times10^{-5}$
Mars	$5{,}46\times10^7$	$-7{,}84\times10^2$	$1{,}25\times10^{-5}$
Mercury	$7{,}73\times10^7$	$-2{,}85\times10^2$	$4{,}56\times10^{-6}$

Table 1

Where we can see that we can also neglect the potential differences they generate on the terrestrial field for the purposes of the GNSS system.

As a point of interest, although we can neglect their gravitational effects regarding the calculation of the temporal correction to be applied to the satellites, we must take them into account in orbital calculations; in fact, in GNSS systems in use today, they are considered in the propagation of ephemerides.

3.4. Total Relativistic Correction

Since the proper time for satellites in orbit advances faster due to their gravitational potential (the offset will have a positive sign) but lags due to their velocity (offset with a negative sign), the total necessary correction derived from relativistic effects for weak fields and circular orbits is:

(3.34)

$$\Delta\tau_{\text{total}} = \Delta\tau_{\text{gravity}} - \Delta\tau_{\text{velocity}} \approx \left(\frac{r_s}{2r} - \frac{r_s}{2(r+\Delta r)}\right) - \frac{r_s}{4(r+\Delta r)}$$

which can be expressed as:

(3.35)

$$\Delta\tau_{\text{total}} \approx \frac{r_s}{2r} - \frac{3r_s}{4(r+\Delta r)} = \frac{GM}{rc^2} - \frac{3GM}{2(r+\Delta r)c^2}$$

Thus, evaluating the expression for the satellites relative to ground observers at potential V_t, we have:

(3.36)

$$\boxed{\Delta\tau_{\text{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2r_{sat}c^2}}$$

which is the expression that will provide the offset to be corrected in the satellites to avoid errors induced by relativistic effects.

3.5. Sensitivity Analysis

We can perform a sensitivity analysis for all significant parameters to see how their variation influences the measurement of the passage of time in the case of two mobiles within a radial field, particularly in the time calculations required for geolocation systems.

3.5.1. Sensitivity Analysis for Velocity Difference

Since the orbit of the satellites remains constant, we can assume that their orbital velocity is also constant. Similarly, we cannot expect changes in the Earth's rotational motion; therefore, the velocity difference between the satellites and the ground observers will remain relatively constant, in addition to all previous considerations regarding the ECI reference system and software correction.

Observers on the surface moving at high speeds may occur, but those corrections fall within the interferences to be analyzed in other software corrections that lie outside the scope of this document.

3.5.2. Sensitivity Analysis for Potential Difference

Recovering the previous expressions after defining proper time τ such that $ds^2 = -c^2 d\tau^2$, and therefore:

(3.37)

$$\mathrm{d}\tau^2 = \left(1 - \frac{r_s}{r}\right) \mathrm{d}t^2 \Rightarrow \frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

we can recover the concept of compactness u and define the previous relationship as:

(3.38)

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{(1 - \mathrm{u})}$$

which provides the differential variation in proper time as a function of the compactness of the radial field, taking the form shown in next figure (figure 9)

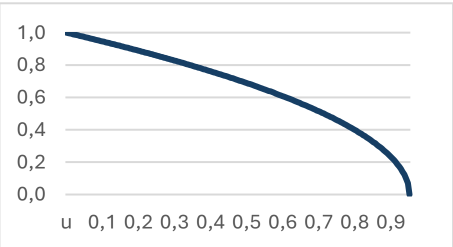


Figure 9

We can see that at points very far from the Schwarzschild radius, where compactness tends toward zero, time dilation is not appreciable, while for objects near the Schwarzschild radius, time tends to dilate infinitely.

3.5.3. Application to the Case of Geolocation Satellites

For the sensitivity analysis in satellite geolocation systems, we find ourselves in a case where the field is weak (compactness can be considered zero) and the relative velocity difference is constant; therefore, the following expression is completely valid:

(3.39)

$$\Delta\tau_{\mathrm{x}} = \left(\frac{r_s}{2(\mathrm{r_t} + \mathrm{x})} - \frac{r_s}{2\big((\mathrm{r_t} + \mathrm{x}) + h\big)}\right)$$

and it will provide, letting x be the altitude above the Earth's surface, the variation in the measurement of proper time relative to the surface.

Assuming we move within the vicinity of the surface $x \ll r_t$ we have $r_t + x \approx r_t$; therefore, we can affirm that variation in altitude does not alter the measurement in positioning systems as long as we remain in the vicinity of the surface.

3.6. Verification of Previous Derivations in Operational Systems

We want to validate the previous derivations by introducing specific values for systems already in use on Earth. Thus, for GPS specifically, according to 3.36:

$$\Delta\tau_{\mathrm{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2\mathrm{r_{sat}}\, c^2}$$

With values

(#4) $\mathrm{V_t} = 6\,263\,686.085 \times 10\ \mathrm{m^2/s^2}$

(#6) $c = 299\,792\,458\ \mathrm{m/s}$

(#4) $GM = 3\,986\,005 \times 10^8\ \mathrm{m^3/s^2}$

(#6) $\mathrm{r_{sat}} = 26\,559\,710\ \mathrm{m}$ (real elliptical orbit semi-major axis)

the formula to apply is:

$$\Delta\tau_{\mathrm{total}} \approx \frac{6\,263\,686.0850 \times 10}{299\,792\,458^2} - \frac{3 \times 3\,986\,005 \times 10^8}{2 \times 26\,559\,710 \times 299\,792\,458^2}$$

$$= 6.9693 \times 10^{-10} - 2.5047 \times 10^{-10}$$

$$= 4.4646 \times 10^{-10}$$

Consequently, the proper time of the satellites leads relative to the Earth's surface by 4.4646×10^{-10} s for every second that passes on the surface. That is, it will daily lead by:

$\Delta\tau_{\text{daily}} \approx$

$(4.4646 \times 10^{-10} \times 3600 \times 24)\ s/day$

$= 38.574 \times 10^{-6}\ s/day = 38.5\ \mu s/day$

This is exactly the correction applied in real GPS according to its ICD (#6, 3.3.1.1 Frequency Plan) , which states that if the nominal frequency of GPS clocks on the surface is 10.23 MHz, the frequency on the satellites must be adjusted to:

$\mathrm{f}' = (1 - \Delta\tau_{total}) \times \mathrm{f}$

$= (1 - 4.4647 \times 10^{-10}) \times 10.23\ \mathrm{MHz}$

$= 10.2299999954326\ \mathrm{MHz}$

This adjustment is introduced into GPS clocks before launch. It has been verified that the expression also returns correct values for other systems, although in Galileo the time correction is performed via software and not by modifying the onboard clock offset. The results are shown in the following table (Table 2):

GPS	
r_{sat}	26 559 710 (#6)
$\Delta\tau_{\text{total}}$ calculated	4.4646×10^{-10}
$\Delta\tau_{\text{total}}$ sources	4.4647×10^{-10} (#6)
$\Delta\tau_{\text{daily}}$	38.5 μs/day
GLONASS	
r_{sat}	25 478 136 (#7)
$\Delta\tau_{\text{total}}$ calculated	4.3581×10^{-10}
$\Delta\tau_{\text{total}}$ sources	4.36×10^{-10} (#7)
$\Delta\tau_{\text{daily}}$	37.6 μs/day
Galileo	
r_{sat}	29 601 297 (#8)
$\Delta\tau_{\text{total}}$ calculated	4.7216×10^{-10}
$\Delta\tau_{\text{total}}$ sources	4.7219×10^{-10} (#3)
$\Delta\tau_{\text{daily}}$	40.7 μs/day

Table 2

As a point of interest and proof of the previous conclusions regarding the sensitivity of the GPS system, the error introduced if we wanted to use it at an altitude of 10 km (the flight altitude of commercial aviation) would be:

$$\Delta\tau_x = \left(\frac{r_s}{2(r_t + x)} - \frac{r_s}{2((r_t + x) + h)}\right)$$
$$= \frac{8.8701 \times 10^{-3}}{2}\left(\frac{1}{(6\,371 + 10) \times 10^3} - \frac{1}{(6\,371 + 10 + 20\,200) \times 10^3}\right)$$
$$= 5.2818 \times 10^{-10}$$

Which, as we see, is equivalent to the calculation on the surface by $\frac{5.2818\times10^{-10}}{5.2922 \times 10^{-10}} = 99.80\%$, i.e., to the relativity correction already predetermined in the GPS system. Also, as a point of interest, if proper time were not corrected in the satellites, the accumulated daily error in each distance measured relative to GPS satellites would be:

$\Delta\mathrm{d}_{\text{daily}} \approx 38.5145 \times 10^{-6}\ s/day \times \mathrm{c}$

$= 11\,546.35\ m/day$ $=$ 11.5 km each day

4. Annexes and Derivations

In this section, I provide some general notes and derivations regarding analytical mechanics and general relativity that are relevant to this work, as well as others directly related to satellite navigation.

4.1. Orbits in GNSS Satellites

GNSS satellites follow free-fall orbits with typically low eccentricity ($\lesssim 0.02$) allowing us to treat them as perfect circles for the purpose of computing linear velocity, i.e., the tangential component of the trajectory. This is not the case for their ephemerides, where position calculations would be incorrect if the exact eccentricity were not implemented.

Since one of the fundamental design parameters is to cover as much of the Earth's surface as possible with precision while minimizing other costs—such as the number of satellites required—and since frequent updates from the ground are necessary, medium Earth orbits are defined below approximately 35,000 km altitude, around 20,000 km. This avoids the inner Van Allen belt, which would destroy electronics and atomic clocks.

We know that the tangential velocity is $v = \sqrt{\frac{GM}{r}}$ and can also be expressed as

$v = \omega \times r$. We also know that ω is the angular distance traveled per unit time, so for a time T in which a full circular trajectory is completed:

(3.40)

$$v = \omega\, r = \frac{2\pi}{T} r$$

Equating both expressions:

(3.41)

$$v = \sqrt{\frac{GM}{r}} = \omega \times r = \frac{2\pi}{T} r \;\Rightarrow\; r = \sqrt[3]{\frac{GMT^2}{(4\pi^2)}}$$

This gives the relationship between the radius of a stable perfectly circular orbit and its period. For current global GNSS systems, we obtain (Table 3):

GNSS	Period (T)	r_{sat} (km)	h (km)
GPS (#6)	11h 58min	26,560	20,182
GLONASS (#7)	11h 15min	25,510	19,140
Galileo (#8)	14h 05min	29,593	23,222
BeiDou (#9)	12h 53min	27,906	21,535

Table 3

These orbits also satisfy other design constraints beyond the absence of drag required for stable circular orbits with constant angular momentum. Using higher radii would require higher signal power, and the cost of placing satellites in such orbits would also be greater.

4.2. Treatment of the Relativistic Correction in Technical Publications (Virial)

We observe that the work carried out in this document is a sufficiently accurate approximation to the published values for GNSS systems in use.

A notable point is the formula used in Ashby (#2, equation 35) , which is identical to the one used here, substituting the radius with the semi-major axis of the ellipse and using properties of the WGS84 geoid:

$$\Delta\tau = \frac{3GM_{\mathrm{E}}}{2ac^2} + \frac{\Phi_0}{c^2}$$

This corresponds to the formulation derived in (3.36), where we changed signs and assigned opposite signs to the velocity and gravitational corrections:

$$\Delta\tau_{\mathrm{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2\mathrm{r}_{\mathrm{sat}}\, c^2}$$

It is particularly notable that the correction is the same even though Ashby uses conservation of energy in the Keplerian elliptical trajectory, while here we assumed a perfect circular orbit. Ashby uses the real trajectory and works with eccentricity (denoted e, defined as the ratio between the focal distance and the semi-major axis), even if it is minimal (GPS max eccentricity 0.02; GLONASS, Galileo, BeiDou: 0.000–0.001, .#6, #7, #8 and #9).

This coincidence arises from the virial theorem, which states that a time-averaged Keplerian elliptical orbit has the same mean kinetic and potential energy as a perfectly circular orbit whose radius equals the ellipse's semi-major axis.

The analytical mechanics required to incorporate eccentricity has been omitted, but any interested reader is encouraged to work through it, as well as the Sagnac correction (much simpler), to better understand relativistic effects in GNSS systems, both at the satellite and ground-receiver level.

4.3. Some Taylor Approximations

In works of special and general relativity, we will frequently encounter two forms for which it will be useful to employ the following Taylor approximations:

$$f(x) = \sqrt{1-x} \approx 1 - \tfrac{1}{2}x \qquad \text{valid for } |x| \ll 1$$

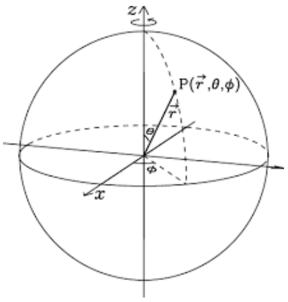
$$f(x) = (1+x)^n \approx 1 + nx \qquad \text{valid for} |x| \ll 1$$

In general relativity, x appears as the compactness $u = \frac{r_s}{r}$ and can be approximated to zero for weak fields (having an order of magnitude u $\sim\ 10^{-27-n+b}$ where n is the order of magnitude of the radius and b is that of the mass), making the expression perfectly valid.

In special relativity, x appears as the ratio $\left(\frac{v}{c}\right)^2$ subtracting in the root of the denominator of the Lorentz transformation, which makes the expression perfectly valid for velocities much lower than c .

4.4. Schwarzschild Solution for Static Spherical Symmetries

We are going to demonstrate the Schwarzschild solution starting from the Minkowski metric in spherical coordinates of the form:



$r \in [0, \infty)$
$\theta \in [0, \pi]$
$\phi \in [0, 2\pi)$

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$$

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

$$ds^2 = -c^2dt^2 + dr^2 + r^2d\theta^2 + r^2sin^2\theta d\phi^2$$

Which we will see frequently written in the following way to summarize the angular coordinates:

$$\mathrm{ds}^2 = -c^2\mathrm{dt}^2 + \mathrm{dr}^2 + r^2\,(\mathrm{d}\theta^2 + \sin^2\theta d\phi^2) = -c^2\mathrm{dt}^2 + \mathrm{dr}^2 + r^2\mathrm{d}\Omega^2$$

And we will formulate a similar spherical symmetry where all free variables are expressed as functions $f_i(r)$ of the radius and only of the radius (the expression for the angular coordinate remains unchanged to maintain the symmetry):

$$\mathrm{ds}^2 = -\mathrm{c}^2 f_t(r)\mathrm{dt}^2 + f_r(r)\mathrm{dr}^2 + r^2\mathrm{d}\Omega^2$$

It will prove convenient to write the functions $f_i(r)$ from the previous expression as exponential functions multiplied by 2 to simplify all subsequent calculations, and therefore:

$$ds^2 = -c^2e^{2f_t(r)}\mathrm{dt}^2 + e^{2f_r(r)}\mathrm{dr}^2 + r^2\,\mathrm{d}\Omega^2$$

We want to calculate the functions $f_i(r)$, i.e., the terms $e^{f_i(r)}$ that make the spherical metric satisfy Einstein's field equation in a vacuum ($T_{ab} = 0$) and without a cosmological constant Λ (because we are calculating the simple solution with an asymptotically flat spacetime):

$$R_{ab} - \frac{1}{2}Rg_{ab} = \frac{8\pi G}{c^4}T_{ab} = 0$$

Where we see that $R_{ab} = 0$ is a necessary condition to satisfy the equation (if $R_{ab} = 0$ its scalar R will also be null for any metric g_{ab}). Therefore, the problem is limited to finding the R_{ab} tensor for the postulated metric and knowing the value that the functions $f_i(r)$ must satisfy so that $R_{ab} = 0$

4.4.1. Calculation of Christoffel Symbols

We will need to find all the Christoffel symbols of this metric necessary to calculate the components of the Ricci tensor to be equated to zero, and we will do so through the principle of least action.

For this, we will define a Lagrangian of the form $L = \left(\frac{ds}{d\lambda}\right)^2$ with an affine parameter λ in which L remains constant $\frac{\partial L}{\partial \lambda} = 0$. Expressing the metric in tensorial notation as $\mathrm{ds}^2 = g_{ab}dx^a dx^b$ and using the notation $\dot{x} = \frac{dx}{d\lambda}$ the Lagrangian can be written as $\mathrm{L} = \left(\frac{ds}{d\lambda}\right)^2 = g_{ab}\dot{x}^a\dot{x}^b$

This Lagrangian applied to the metric in the exponential form we had proposed results in:

$$\mathrm{L} = -\mathrm{c}^2 e^{2f_t(r)}\dot{t}^2 + e^{2f_r(r)}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2$$

Where we have expanded the angular component again to be able to find the Christoffel symbols associated with θ and ϕ. We will expand all the terms of the Euler-Lagrange least-action expression with that Lagrangian:

$$\frac{d}{d\lambda}\left(\frac{\partial \mathrm{L}}{\partial \dot{x}^a}\right) - \frac{\partial \mathrm{L}}{\partial x^a} = 0$$

and then we will compare the resulting terms with the geodesic equation in canonical form to obtain the Γ^a for each coordinate:

$$\ddot{x^a} + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$$

Expanding those Γ^a_{bc} for our four-dimensional space for each coordinate x^a according to the expressions of all the other possible pairs of coordinates x^a and x^b

of the metric, the comparison of terms will be immediate.

Derivation for the t coordinate

$$L = -c^2 e^{2f_t(r)}\dot{t}^2 + e^{2f_r(r)}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2$$

Recall that all variables are independent and their derivatives are as well (state space):

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{t}}\right) = \frac{d}{d\lambda}\left(-2c^2 e^{2f_t(r)}\dot{t}\right)$$

$$= \frac{d}{d\lambda}\left(-2c^2 e^{2f_t(r)}\right)\dot{t} - 2c^2 e^{2f_t(r)}\ddot{t}$$

Applying the chain rule $\frac{d}{d\lambda}\left[f\left(r(\lambda)\right)\right] = \frac{df}{dr}\cdot\frac{dr}{d\lambda}$ which we can write as $\frac{df}{dr}\dot{r} = f'\dot{r}$ but not as $\dot{f}\dot{r}$ because f is not differentiated with respect to the affine parameter

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{t}}\right) = \left[-4c^2 f_t'(r)e^{2f_t(r)}\dot{r}\right]\dot{t} - 2c^2 e^{2f_t(r)}\ddot{t}$$

$$\frac{\partial L}{\partial t} = 0$$

Therefore, the geodesic equation can be written as:

$$\left[-4f_t'(r)c^2 e^{2f_t(r)}\dot{r}\right]\dot{t} - 2c^2 e^{2f_t(r)}\ddot{t} = 0$$

Dividing by $-2c^2 e^{2f_t(r)}$, we find that:

$\ddot{t} + 2f_t'(r)\,\dot{r}\,\dot{t} = 0$ which, compared with $\ddot{x^a} + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$

indicates that $\Gamma^t_{rt} = \Gamma^t_{tr} = f_t'(r)$ and the rest of the Christoffel symbols in t are zero.

Conservation of Energy

It is important to note here that $\frac{\partial L}{\partial t} = 0$, so the Lagrangian is conserved in t for any affine parameter and that from said conservation it also follows that: $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{t}}\right) = 0$ where $\frac{\partial L}{\partial \dot{t}} = -2c^2 e^{2f_t(r)}\dot{t}$

Therefore $-2c^2 e^{2f_t(r)}\dot{t}$ is a constant with respect to any affine parameter. We will call this constant magnitude energy per unit mass ϵ or specific energy, and its constancy, conservation of energy. And for its general use, we will eliminate all constants whose derivative is zero so as not to incorrectly dimension the equations in which we use it, in the form:

$$\epsilon = e^{2f_t(r)}\dot{t} \Leftrightarrow \frac{\partial L}{\partial t} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{t}} = \epsilon$$

This conservation is maintained under coordinate changes in ϕ and t, although not in coordinate changes in r or θ.

Derivation for the r coordinate

$$L = -c^2 e^{2f_t(r)}\dot{t}^2 + e^{2f_r(r)}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2$$

Recall that all variables are independent and their derivatives are as well (state space):

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{r}}\right) = \frac{d}{d\lambda}\left(2e^{2f_r(r)}\dot{r}\right) = \frac{d}{d\lambda}\left(2e^{2f_r(r)}\right)\dot{r} + 2e^{2f_r(r)}\ddot{r} =$$

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{r}}\right) = 4f_r'(r)e^{2f_r(r)}\dot{r}^2 + 2e^{2f_r(r)}\ddot{r}$$

$$\frac{\partial L}{\partial r} = 2r\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) - 2c^2 f_t'(r)\,e^{2f_t(r)}\dot{t}^2 + 2f_r'(r)\,e^{2f_r(r)}\dot{r}^2$$

Therefore, the geodesic equation can be written as:

$$4f_r'(r)e^{2f_r(r)}\dot{r}^2 + 2e^{2f_r(r)}\ddot{r} - \left[2r\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) - 2c^2 f_t'(r)\,e^{2f_t(r)}\dot{t}^2 + 2f_r'(r)\,e^{2f_r(r)}\dot{r}^2\right] = 0$$

$$4f_r'(r)e^{2f_r(r)}\dot{r}^2 + 2e^{2f_r(r)}\ddot{r} - 2r\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + 2c^2 f_t'(r)\,e^{2f_t(r)}\dot{t}^2 - 2f_r'(r)e^{2f_r(r)}\dot{r}^2 = 0$$

$$2e^{2f_r(r)}\ddot{r} + 4f_r'(r)e^{2f_r(r)}\dot{r}^2 + - 2r\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + 2c^2 f_t'(r)\,e^{2f_t(r)}\dot{t}^2 - 2f_r'(r)e^{2f_r(r)}\dot{r}^2 = 0$$

$$2e^{2f_r(r)}\ddot{r} + 2f_r'(r)e^{2f_r(r)}\dot{r}^2 + - 2r\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + 2c^2 f_t'(r)\,e^{2f_t(r)}\dot{t}^2 = 0$$

Dividing by $2e^{2f_r(r)}$, we find that:

$$\ddot{r} + f_r'(r)\,\dot{r}^2 - r\left(e^{-2f_r(r)}\right)\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + c^2 f_t'(r)\,e^{2f_t(r)-2f_r(r)}\dot{t}^2 = 0$$

which, compared with $\ddot{x^a} + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$ indicates that

$$\Gamma^r_{rr} = f_r'(r)$$

$$\Gamma^r_{\theta\theta} = -r\,e^{-2f_r(r)}$$

$$\Gamma^r_{\phi\phi} = -r\sin^2\theta\;e^{-2f_r(r)}$$

$\Gamma^r_{tt} = c^2 f_t'(r)\, e^{2f_t(r)-2f_r(r)}$

and the rest of the Christoffel symbols in r are zero.

Derivation for the θ coordinate

$$L = -c^2 e^{2f_t(r)}\dot{t}^2 + e^{2f_r(r)}\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2\theta\, \dot{\phi}^2$$

Recall that all variables are independent and their derivatives are as well (state space):

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \frac{d}{d\lambda}(2r^2\,\dot{\theta}) = \frac{d}{d\lambda}(2r^2)\dot{\theta} + 2r^2\ddot{\theta} = 4r\,\dot{r}\dot{\theta} + 2r^2\ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = 2r^2 \sin\theta\cos\theta\,\dot{\phi}^2 = r^2 \sin 2\theta\ \dot{\phi}^2$$

Therefore, the geodesic equation can be written as:

$$4r\dot{r}\dot{\theta} + 2r^2\ddot{\theta} - r^2 \sin 2\theta\ \dot{\phi}^2 = 0$$

Dividing by $2r^2$ we have

$$\ddot{\theta} + (2/r)\dot{r}\dot{\theta} - (1/2)\sin 2\theta\,\dot{\phi}^2 = 0$$

which compared with $\ddot{x^a} + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$ indicates that

$$\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}$$

$$\Gamma^\theta_{\phi\phi} = -(1/2)\sin 2\theta$$

and the rest of the Christoffel symbols in θ are zero.

Derivation for the ϕ coordinate

$$L = -c^2 e^{2f_t(r)}\dot{t}^2 + e^{2f_r(r)}\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2\theta\, \dot{\phi}^2$$

Recall that all variables are independent and their derivatives are as well (state space):

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = \frac{d}{d\lambda}(2r^2 \sin^2\theta\dot{\phi}) = \frac{d}{d\lambda}(2r^2\sin^2\theta)\dot{\phi} + 2r^2\sin^2\theta\ddot{\phi}$$

$$=(4r^2 \sin\theta\cos\theta\,\dot{\theta} + 4r\dot{r}\ \sin^2\theta)\dot{\phi} + 2r^2\sin^2\theta\ddot{\phi} = 4r^2\sin\theta\cos\theta\,\dot{\theta}\dot{\phi} + 4r\dot{r}\ \sin^2\theta\dot{\phi} + 2r^2\sin^2\theta\ddot{\phi}$$

$$\frac{\partial L}{\partial \phi} = 0$$

Therefore, the geodesic equation can be written as:

$$4r^2 \sin\theta\cos\theta\,\dot{\theta}\dot{\phi} + 4r\dot{r}\ \sin^2\theta\dot{\phi} + 2r^2\sin^2\theta\ddot{\phi} = 0$$

Dividing by $2r^2 \sin^2\theta$ we have

$$\ddot{\phi} + 2\cot\theta\,\dot{\theta}\dot{\phi} + 2(1/r)\dot{r}\,\dot{\phi} = 0$$

which compared with $\ddot{x^a} + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$ indicates that

$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta$$

$$\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = \frac{1}{r}$$

and the rest of the Christoffel symbols in ϕ are zero.

Conservation of Angular Momentum

It is important to note here also that $\frac{\partial L}{\partial \phi} = 0$ so the Lagrangian is conserved in ϕ for any affine parameter, and from said conservation it also follows that:

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = 0 \text{ where } \frac{\partial L}{\partial \dot{\phi}} = 2r^2\sin^2\theta\dot{\phi}$$

Therefore $2r^2\sin^2\theta\dot{\phi}$ is a constant with respect to any affine parameter. We will call this constant magnitude angular momentum per unit mass or specific momentum j (to distinguish it from the Lagrangian L), and its constancy, conservation of angular momentum. And for its general use, we will eliminate all constants whose derivative is zero so as not to incorrectly dimension the equations in which we use it, in the form:

$$j = r^2\sin^2\theta\,\dot{\phi} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{\phi}} = j$$

This conservation is maintained under coordinate changes in ϕ and t, although not in coordinate changes in r or θ.

4.4.2. Calculation of the Ricci Tensor Components R_{ab}

We are now in a position to obtain the Ricci components and equate them to zero to obtain the exponential functions of the proposed metric with the following summary table of non-zero Christoffel symbols:

Christoffel Symbols in Schwarzschild		
$\Gamma^t_{rt} = \Gamma^t_{tr} = f_t'(r)$	$\Gamma^r_{\phi\phi} = -r\sin^2\theta\ e^{-2f_r(r)}$	$\Gamma^r_{tt} = c^2 f_t'(r)\, e^{2f_t(r)-2f_r(r)}$
$\Gamma^r_{rr} = f_r'(r)$	$\Gamma^r_{\theta\theta} = -r\, e^{-2f_r(r)}$	$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta$
$\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}$	$\Gamma^\theta_{\phi\phi} = -(1/2)\sin 2\theta$	$\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = \frac{1}{r}$

We will derivate only one of the tensor components to show the complexity of the calculation and for this reason, this will be the only explicit component of R_{ab} calculated in this document. For the complete Schwarzschild solution, we will directly introduce the values of R_{ab} that can be obtained from multiple sources, as this calculation is frequent in general relativity problems for the postulated metric.

Particular calculation for R_{tt}

$$R_{tt} = \partial_r \Gamma^r_{tt} + \Gamma^r_{tt}\left(\Gamma^r_{rr} + \Gamma^\theta_{r\theta} + \Gamma^\phi_{r\phi}\right) - \Gamma^t_{tr}\Gamma^r_{tt}$$

We find that the only previous partial derivative corresponds to:

$$\partial_r \Gamma^r_{tt} = \partial_r\left(c^2 f_t'(r) e^{2f_t(r)-2f_r(r)}\right) = c^2\left[f_t''(r) + 2\left(f_t'(r)\right)^2 - 2f_t'(r)f_r'(r)\right] e^{2f_t(r)-2f_r(r)}$$

Substituting and expanding we have

$$R_{tt} = c^2\left[f_t''(r) + 2\left(f_t'(r)\right)^2 - 2f_t'(r)f_r'(r)\right] e^{2f_t(r)-2f_r(r)} + c^2 f_t'(r)\, e^{2f_t(r)-2f_r(r)}\left[\left(f_r'(r)\right) + \frac{1}{r} + \frac{1}{r}\right] - c^2\, f_t'(r)^2\, e^{2f_t(r)-2f_r(r)}$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + 2\left(f_t'(r)\right)^2 - 2f_t'(r)f_r'(r) + f_t'(r)\left(\left(f_r'(r)\right) + \frac{1}{r} + \frac{1}{r}\right) - f_t'(r)^2\right]$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + \left(f_t'(r)\right)^2 - 2f_t'(r)f_r'(r) + f_t'(r)\left(\frac{rf_r'(r)+2}{r}\right)\right]$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - 2f_t'(r)f_r'(r) + f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right]$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right]$$

Together with the rest of the non-zero components for the Ricci tensor that we can extract from different sources (#1, equations 5.14), we have

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right]$$

$$R_{rr} = f_t''(r) + \left(f_t'(r)\right)^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}$$

$$R_{\theta\theta} = e^{-2f_r(r)}\left[1 + r\left(f_t'(r) - f_r'(r)\right)\right] - 1$$

$$R_{\phi\phi} = \sin^2\theta \cdot R_{\theta\theta} = \sin^2\theta\left(e^{-2f_r(r)}\left[1 + r\left(f_t'(r) - f_r'(r)\right)\right] - 1\right)$$

Where it is observed that all cross components disappear and that those that are not null depend on the radius and only on the radius, which is consistent with the initial assumption of spherical symmetry with a diagonal metric dependent on the radius and only on the radius.

4.4.3. Integration of R_{ab} in the EFE

4.4.3.1. Individual Components of R_{ab} in the EFE

Now we will equate all non-zero components of $R_{ab} = 0$ to obtain the sought functions $f_r(r)$ and $f_t(r)$

$$R_{tt} = 0 \Rightarrow c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right] = 0$$

$$R_{rr} = 0 \Rightarrow f_t''(r) + \left(f_t'(r)\right)^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r} = 0$$

$$R_{\theta\theta} = 0 \Rightarrow e^{-2f_r(r)}\left[1 + r\left(f_t'(r) - f_r'(r)\right)\right] - 1 = 0$$

$R_{\phi\phi} = 0 \Rightarrow \sin^2\theta \cdot R_{\theta\theta} = \sin^2\theta \left(e^{-2f_r(r)}\left[1 + r\left(f_t'(r) - f_r'(r)\right)\right] - 1\right) = 0$

Of all the previous coupled equations, we will first choose $R_{\theta\theta} = 0$ because it is the one that implies immediate consequences for solving the system.

$R_{\theta\theta} = 0 \Rightarrow e^{-2f_r(r)}\left[1 + r\left(f_t'(r) - f_r'(r)\right)\right] = 1$

$\Rightarrow \left[1 + r\left(f_t'(r) - f_r'(r)\right)\right] = e^{2f_r(r)}$

$\Rightarrow r\left(f_t'(r) - f_r'(r)\right) = e^{2f_r(r)} - 1$

$(R_{\theta\theta} = 0) \Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)} - 1}{r}$

4.4.3.2. Comparison of R_{tt} and R_{rr} components

Showing all components in a single expression we have

$R_{tt} + R_{rr} + R_{\theta\theta} + R_{\phi\phi} = 0$

or normalizing in the scalar

$$\Rightarrow \mathrm{R} = g^{ab}R_{ab} = g^{tt}R_{tt} + g^{rr}R_{rr} + g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} + \frac{R_{\theta\theta}}{g_{\theta\theta}} + \frac{R_{\phi\phi}}{g_{\phi\phi}} = 0$$

Having proposed a spherical symmetry where $g_{\theta\theta}$ *and* $g_{\phi\phi}$ do not contain the functions $f_t(r)\ y\ f_r(r)$ that we want to deduce, we will not be able to use their components, but we can use those of t and r, in the form

$$\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = g^{tt}R_{tt} + g^{rr}R_{rr} = 0$$

And substituting the values of the individual expressions into this expression

$$\frac{c^2 e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right]}{-c^2 e^{2f_t(r)}} + \frac{R_{rr}f_t''(r) + \left(f_t'(r)\right)^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}}{e^{2f_r(r)}} = 0$$

$$\frac{-e^{2f_t(r)-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right]}{e^{2f_t(r)}} + \frac{f_t''(r) + \left(f_t'(r)\right)^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}}{e^{2f_r(r)}} = 0$$

$$-e^{-2f_r(r)}\left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r)\right] - e^{2f_r(r)}\left[f_t''(r) + \left(f_t'(r)\right)^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}\right] = 0$$

And canceling term by term in $e^{-2f_r(r)}$

$$-e^{-2f_r(r)}\left[+\frac{2f_t'(r)}{r}\right] - e^{2f_r(r)}\left[-\frac{2f_r'(r)}{r}\right] = 0$$

And since the exponential can never be zero, we find that:

$$\frac{2}{r}[f_t'(r) + f_r'(r)] = 0$$

and therefore:

$$f_t'(r) + f_r'(r) = 0$$

$$\left(\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = 0\right) \Rightarrow f_t'(r) = -f_r'(r)$$

4.4.3.3. Integration of the Previous Derivations

Thus, applying the two expressions we concluded previously, we have:

$(R_{\theta\theta} = 0) \Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)}-1}{r}$ and $\left(\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = 0\right) \Rightarrow f_t'(r) = -f_r'(r)$

$\Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)}-1}{r}$

$\Rightarrow 2\mathrm{r}f_t'(r) = e^{-2f_t(r)} - 1$

which is a first-order ordinary differential equation with solution:

$f_t(r) = \frac{1}{2}\ln\left(1 - \frac{C}{r}\right)$ where C is a constant of integration

And according to our assumption $f_t'(r) = -f_r'(r)$ we find that

$$f_r(r) = -\frac{1}{2}\ln\left(1-\frac{C}{r}\right)$$

It is convenient to note that the previous expressions can be written exponentially as:

$e^{2f_t(r)} = 1-\frac{C}{r}$ and $e^{2f_r(r)} = \frac{1}{1-\frac{C}{r}}$

And therefore, the initial metric can be written as:

$$ds^2 = -c^2 e^{2f_t(r)}dt^2 + e^{2f_r(r)}dr^2 + r^2\,d\Omega^2$$

$$ds^2 = -c^2\left(1-\frac{C}{r}\right)dt^2 + \left(\frac{1}{1-\frac{C}{r}}\right)dr^2 + r^2 d\Omega^2$$

$$ds^2 = -c^2\left(1-\frac{C}{r}\right)dt^2 + \left(1-\frac{C}{r}\right)^{-1}dr^2 + r^2 d\Omega^2$$

Where it remains for us to determine the value of the integration constant C for the conditions of the problem.

4.4.4. Determination of Mass (Newtonian Limit)

To do this, we are going to make three assumptions based on classical mechanics.

The first is to assume that this integration constant C is directly proportional to the mass generating the field, such that: $C \propto M$

The second is the assumption that when r tends to infinity, the metric takes the form of a flat space for that mass, such that:

$$\lim_{r\to\infty}\left(1-\frac{C \propto M}{r}\right) = 1 \quad , \lim_{r\to\infty}\left(1-\frac{C \propto M}{r}\right)^{-1} = 1$$

And the third is that at infinite r, our equations must behave as in classical mechanics (Newtonian limit); that is, we will impose those conditions on the equations we have derived.

If we start from the geodesic equation for the radial component derived from the proposed solution particularized for r, according to its non-zero Christoffel symbols:

$\ddot{r} + \Gamma^r_{rr}\dot{r}\dot{r} + \Gamma^r_{\theta\theta}\dot{\theta}\dot{\theta} + \Gamma^r_{\phi\phi}\dot{\phi}\dot{\phi} + \Gamma^r_{tt}\dot{t}\dot{t} = 0$ with the affine parameter being the proper time τ

In the Newtonian limit, velocities are much lower than that of light, so any component that has a spatial derivative is negligible compared to the other time-dependent components. On the other hand, when the affine parameter is the proper time, its derivative can be equated to 1 at infinity.

$\Gamma^r_{rr}\dot{r}\dot{r} + \Gamma^r_{\theta\theta}\dot{\theta}\dot{\theta} + \Gamma^r_{\phi\phi}\dot{\phi}\dot{\phi} \approx 0$ and $\dot{t} \approx 1$

$$\ddot{r} + \Gamma^r_{tt}\dot{t}\dot{t} \approx 0 \Rightarrow \ddot{r} + \Gamma^r_{tt} \approx 0$$

Now we bring the expression of Γ^r_{tt} from the metric:

$$\Gamma^r_{tt} = \frac{1}{2}g^{rr}(\partial_t g_{rt} + \partial_t g_{tr} - \partial_r g_{tt})$$

Given that we are working in a static solution, whose metric does not depend on time, all partial derivatives with respect to time disappear:

$$\Gamma^r_{tt} = \frac{1}{2}g^{rr}(-\partial_r g_{tt})$$

And since we have $g^{rr} = 1/g_{rr}$

$$\Gamma^r_{tt} = -\frac{1}{2g_{rr}}(\partial_r g_{tt})$$

And we recall that:

$g_{tt} = -c^2\left(1-\frac{C}{r}\right)$ and $g_{rr} = \left(1-\frac{C}{r}\right)^{-1}$

$$\partial_r g_{tt} = \partial_r\left[-\left(1-\frac{C}{r}\right)\right] = c^2\frac{C}{r^2}$$

Thus

$$\Gamma^r_{tt} = -\frac{1}{2g_{rr}}(\partial_r g_{tt}) = -\frac{1}{2g_{rr}}c^2\frac{C}{r^2} = -c^2\frac{C}{2r^2}\left(1-\frac{C}{r}\right)$$

Returning to the geodesic equation for the radius:

$\ddot{r} - \frac{C}{2r^2}c^2\left(1-\frac{C}{r}\right) \approx 0$ and recalling the weak-field limit $\lim_{r\to\infty}\left(1-\frac{C \propto M}{r}\right) = 1$ we find that

$$\ddot{r} - \frac{C}{2r^2}[c^2] \approx 0 \Rightarrow \ddot{r} \approx \frac{C}{2r^2}[c^2]$$

comparing with Newtonian acceleration, $a = \frac{GM}{r^2}$, in the limit when r tends to infinity $\ddot{r} = a \Rightarrow \frac{C}{2r^2}[c^2] = \frac{GM}{r^2}$ so $C = \frac{2GM}{c^2}$ which from now on we will call the Schwarzschild radius and denote as $r_s = \frac{2GM}{c^2}$

And substituting into the metric:

$$ds^2 = -c^2\left(1-\frac{r_s}{r}\right)dt^2 + \left(1-\frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \text{ where } r_s = \frac{2GM}{c^2}$$

Which is the common way to show the Schwarzschild metric.

4.4.5. Useful Evaluations on the Schwarzschild Radius

Working with the previous expression for r_s we find that by introducing orders of magnitude for the fundamental constants:

$$r_s = \frac{2GM}{c^2} = \frac{2G}{c^2}M \Rightarrow r_s \sim \frac{10^{-11}}{(10^8)^2}M \Rightarrow r_s \sim \frac{10^{-11}}{10^{16}}M \Rightarrow r_s \sim 10^{-27}M$$

And therefore, in orders of magnitude:

$$r_s \sim 10^{-27}\mathrm{M}$$

We shall call the ratio between r_s *and* M the Schwarzschild radius per unit mass and denote it as r_m. We thus see that $r_m \sim 10^{-27}$ for any mass M.

We find, therefore, that the ratio $\frac{r_s}{r}$, which we shall call compactness u, is a multiple of r_m thus, for radius orders n , the order of compactness will be 10^{-27-n} times the order of mass M.

That is, calling u the compactness:

$$\mathrm{u} = \frac{r_s}{r} = M\frac{r_m}{r}$$

And with $\mathrm{r} \sim \mathrm{n}$, $\mathrm{M} \sim \mathrm{b}$ and $r_m \sim 10^{-27} \Rightarrow \mathrm{u} \sim 10^{-27-n+b}$

Where we see that the order of the radius n and that of the mass M must be significant to alter the compactness: only high masses and small radii will raise the order of magnitude of the compactness.

Being the expression of the metric:

$$ds^2 = -c^2\left(1-\frac{r_s}{r}\right)dt^2 + \left(1-\frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Now written in terms of compactness:

$$ds^2 = -c^2(1-u)dt^2 + (1-\mathrm{u})^{-1}dr^2 + r^2 d\Omega^2$$

We see that for any angular coordinate, at any radius of the spherical symmetry, the order of compactness $\sim 10^{-27-n+b}$ could lead to it being neglected, and therefore:

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$$

Which is exactly the Minkowski metric.

The difference in orders of magnitude $(27 - b + n)$ proves to be a good criterion for neglecting curvature and speaking of weak fields where Newtonian gravity, being simpler, is completely valid. To work with high differences is called the weak-field regime.

As an example, for the Earth $\mathrm{u} \sim 10^{-9}$ imposing order 6 for the radius and 24 for the mass.

4.4.6. Determination of Geodesics

We can now determine the geodesics in a radial gravitational field by recovering the equations for the four dimensions and substituting the found integration constant:

$$\ddot{t} + 2f_t'(r)\,\dot{r}\,\dot{t} = 0$$

$$\ddot{r} + f_r'(r)\,\dot{r}^2 - r\left(e^{-2f_r(r)}\right)\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + c^2 f_t'(r)\, e^{2f_t(r)-2f_r(r)}\dot{t}^2 = 0$$

$$\ddot{\theta} + (2/r)\dot{r}\dot{\theta} - (1/2)\sin 2\theta\,\dot{\phi}^2 = 0$$

$$\ddot{\phi} + 2\cot\theta\,\dot{\theta}\dot{\phi} + 2(1/r)\dot{r}\,\dot{\phi} = 0$$

And summarizing the functions and their derivatives:

$$e^{2f_t(r)} = 1 - \frac{r_s}{r}$$

$$e^{2f_r(r)} = \frac{1}{1-\frac{r_s}{r}}$$

$$f_t'(r) = \frac{r_s}{2r(r-r_s)}$$

$$f_r'(r) = \frac{-r_s}{2r(r-r_s)}$$

It results that the geodesics are:

Geodesics in Schwarzschild	
t	$\ddot{t} + \frac{r_s}{r(r-r_s)}\dot{r}\,\dot{t} = 0$

r	$\ddot{r} - \frac{r_s}{2r(r-r_s)}\dot{r}^2 - r\left(1-\frac{r_s}{r}\right)(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2) + c^2\frac{r_s}{2r(r-r_s)}\dot{t}^2 = 0$
θ	$\ddot{\theta} + (2/r)\dot{r}\dot{\theta} - (1/2)\sin 2\theta\,\dot{\phi}^2 = 0$
ϕ	$\ddot{\phi} + 2\cot\theta\,\dot{\theta}\dot{\phi} + 2(1/r)\dot{r}\,\dot{\phi} = 0$

Recalling also that in the metric, the following cyclic coordinates or conserved magnitudes are observed, where the integration constant can now be substituted:

Specific Energy $\epsilon = e^{2f_t(r)}\dot{t} = \left(1-\frac{r_s}{r}\right)\dot{t} \Leftrightarrow \frac{\partial L}{\partial t} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{t}} = \epsilon$

Angular Momentum $j = r^2\sin^2\theta\,\dot{\phi} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{\phi}} = j$

4.4.7. Extended Application of the Schwarzschild Solution (Birkhoff's Theorem)

Birkhoff's theorem establishes that every solution of the EFE in a vacuum with spherical symmetry is necessarily static and asymptotically flat (its compactness u tends to 0 for $r \gg r_s$), so it coincides with the Schwarzschild metric.

Therefore, the Schwarzschild solution demonstrated above remains valid and equivalent to the one we would obtain by introducing time as a variable in the functions of the metric coordinates.

It also allows us to affirm that although the radius in which the mass is concentrated varies over time, as long as spherical symmetry is maintained, the solution will remain constant; or considering that the symmetry holds, the solution will not vary despite changes in the internal density of, for example, the Sun and the planets of the solar system, including the Earth.

4.4.8. Local Observers (shell)

Now we are going to observe that moving objects never perceive a change in position relative to themselves; they only see their proper time passing. If they also follow a geodesic trajectory, they are in free fall in the radial field that generates it, so by the equivalence principle, they only perceive a flat metric of the form:

$$ds^2 = -c^2 d\tau^2$$

But reference systems that do not experience that free fall do perceive that movement, which must also satisfy the invariance of ds^2 for any observer, so that:

$$ds^2_{falling} = ds^2_{not\ falling}$$

that is

$$-c^2 d\tau^2_{falling} = -c^2\left(1-\frac{r_s}{r}\right)dt^2_{not\ falling} + \left(1-\frac{r_s}{r}\right)^{-1}dr^2_{not\ falling} + r^2 d\Omega^2_{not\ falling}$$

which is the equation that allows us to deduce velocities near the observer in fall, and which we call local observations or shell observations, by considering that they are produced on one of the spherical surfaces of constant r of the metric near the phenomenon to be characterized.

These velocities are not the same as those obtained by dividing the metric, which is asymptotically flat, by dt, which take velocity values valid for coordinate observers—that is, at infinity—but which do not allow plane geometry addition operations impossible with curvature.

In any case, the relationship between velocities of coordinate observers and local or shell observers must follow the time dilation relationship explained here and shown below:

$$\frac{d\tau}{dt} = \sqrt{\left(1-\frac{r_s}{r}\right)}$$

which, as we see, is what allows us—in events far from the Schwarzschild radius—to use the proper time of the falling mobile to characterize the fall from another mobile that is static relative to it.

4.5. Rigid Coordinate Changes Applied to the Principle of Least Action

We are going to leave reflected here some reflections on how rigid coordinate changes affect the principle of least action.

For the Lagrangian $L = \left(\frac{ds}{d\lambda}\right)^2$ and the metric $ds^2 = g_{ab}dx^a dx^b$ expressed in tensorial notation, an affine parameter λ is chosen in which L remains constant $\frac{\partial L}{\partial \lambda} = 0$.

The notation $\dot{x} = \frac{dx}{d\lambda}$ is used with respect to that affine parameter and the change of variable

$x^a = x^a{}_0 + \alpha^a$ is defined where α^a is a constant with respect to the affine parameter (condition 1) and the Lagrangian (condition 2).

It thus results $\dot{x}^a = \dot{x}^a{}_0$ and $\ddot{x}^a = \ddot{x}^a{}_0$ because α^a disappears upon differentiating (condition 1) and that we can continue formulating the Euler-Lagrange equations with this Lagrangian (condition 2), although we cannot introduce it into generic tensorial expressions because we are studying the case of a single change of variable.

For each variable x^a we find that the least action by Euler-Lagrange is of the form

(Ec1) $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) - \frac{\partial L}{\partial x^a} = 0$

and we have to derive

T1 = $\left(\frac{\partial L}{\partial \dot{x}^a}\right)$, T2 = $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ and T3 = $\left(\frac{\partial L}{\partial x^a}\right)$

Derivation of T1

(T1) $\frac{\partial L}{\partial \dot{x}^a} = \frac{\partial}{\partial \dot{x}^a}\left(\frac{ds}{d\lambda}\right)^2 = \frac{\partial}{\partial \dot{x}^a}\left(\frac{1}{d\lambda^2}\, g_{ab} dx^a dx^b\right)$

Considering that all variables in the Lagrangian are independent, this expression will only take a non-zero value for $\dot{x}^a$ and its cross terms:

$$\frac{\partial L}{\partial \dot{x}^a} = 2 g_{ab} \dot{x}^b$$

Recalling that we cannot use $\dot{x}^a = \dot{x}^a{}_0$ from the definition of the coordinate change because it is only valid in the case that a equals b in the expression, therefore:

$$\frac{\partial L}{\partial \dot{x}^a} = 2 g_{ab} \dot{x}^b \neq 2 g_{ab} \dot{x}^b_0$$

That is, a coordinate change in x^a will affect the term $\frac{\partial L}{\partial \dot{x}^a}$ only in $\dot{x}^a$ and its cross terms (when b takes the value a in the previous expression).

Examples of T1 Derivation

In Minkowski in spherical coordinates with $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2$ the term $\frac{\partial L}{\partial \dot{x}^a}$ will only vary under coordinate changes in r and θ but not in ϕ or t.

In Minkowski in Cartesian coordinates $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ the term $\frac{\partial L}{\partial \dot{x}^a}$ will not vary under coordinate changes in any variable.

In Schwarzschild with $ds^2 = -A(r)c^2 dt^2 + B(r) dr^2 + C(r) d\theta^2 + D(r,\theta) d\phi^2$ the term $\frac{\partial L}{\partial \dot{x}^a}$ will vary under coordinate changes in r but not in t, θ or ϕ.

Derivation of T2

(T2) $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = \frac{d}{d\lambda}\left(2 g_{ab} \dot{x}^b\right)$

Considering that we can now find cross-terms where two variables depend on the parameter, we will have to apply the product rule in the form:

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = \frac{d}{d\lambda}\left(2 g_{ab} \dot{x}^b\right) = 2\,\frac{d}{d\lambda}\left(g_{ab} \dot{x}^b\right) = 2\left(\frac{dg_{ab}}{d\lambda}\dot{x}^b + g_{ab}\frac{d\dot{x}^b}{d\lambda}\right) = 2\left(\frac{dg_{ab}}{d\lambda}\dot{x}^b + g_{ab}\ddot{x}^b\right)$$

$$= 2\,\frac{dg_{ab}}{d\lambda}\dot{x}^b + 2\, g_{ab}\ddot{x}^b$$

We decompose the derivative of the metric by introducing a different index μ

$$\frac{dg_{ab}}{d\lambda} = \frac{\partial g_{ab}}{\partial x^\mu}\frac{dx^\mu}{d\lambda} = \frac{\partial g_{ab}}{\partial x^\mu}\dot{x}^\mu = (\partial_\mu g_{ab})\dot{x}^\mu$$

Therefore, returning to the previous expression:

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = 2\,\frac{dg_{ab}}{d\lambda}\dot{x}^b + 2\, g_{ab}\ddot{x}^b = 2\,(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b + 2\, g_{ab}\ddot{x}^b$$

Recalling how the coordinate change is defined, we must analyze how the previous expression is altered when the dummy indexes take the value a, at which point $\dot{x}^a = \dot{x}^a{}_0$ and $\ddot{x}^a = \ddot{x}^a{}_0$

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = 2\,(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b + 2\, g_{ab}\ddot{x}^b$$

The term $g_{ab}\ddot{x}^b$ will only be altered when a = b, where it will be sensitive to the coordinate change in x^a, so we can write that $2\, g_{ab}\ddot{x}^b \neq 2 g_{ab}\ddot{x}^b_0$

The term $(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b$ will only be altered when a = b regardless of whether a = μ, since $\dot{x}^a$ will appear as a common factor of the summation of partial derivatives. Thus we can write that $(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b = 2\,(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b{}_0$

and also observe that we do not violate any explicit rule in the syntax of tensorial calculus (dummy and free indexes are respected).

That is, a coordinate change in x^a will affect the term $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ only in $\dot{x}^a$ and its cross terms (when b takes the value a in the previous expression), just as happened to the term $\frac{\partial L}{\partial \dot{x}^a}$ before taking the total derivative with respect to the parameter.

Examples of T2 Derivations

In Minkowski in spherical coordinates with $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + r^2 sin^2\theta d\phi^2$ the term $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ will only vary under coordinate changes in r and θ but not in ϕ or t.

In Minkowski in Cartesian coordinates $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + dz^2$ the term $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ will not vary under coordinate changes in any variable.

In Schwarzschild with $ds^2 = -A(r)c^2dt^2 + B(r)dr^2 + C(r)d\theta^2 + \mathrm{D}(\mathrm{r},\theta)d\phi^2$ the term $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ will vary under coordinate changes in r but not in t, θ or ϕ.

Derivation of T3

(T3) $\frac{\partial \mathrm{L}}{\partial x^a} = \frac{\partial}{\partial x^a}\left(\frac{ds}{d\lambda}\right)^2 = \frac{\partial}{\partial x^a}\left(\frac{1}{d\lambda^2}\, g_{ab}dx^a dx^b\right)$

Here it is observed that, considering all variables independent, the expression will take a value different from zero in x^a only for its cross terms with the form:

$$\frac{\partial L}{\partial x^a} = \frac{\partial}{\partial x^a}\left(\frac{ds}{d\lambda}\right)^2 = \frac{\partial}{\partial x^a}\left(\frac{1}{d\lambda^2}\, g_{ab}dx^a dx^b\right) = \left(\partial_a g_{b\mu}\right)\dot{x}^b\dot{x}^\mu$$

Which will always be altered for any coordinate change in x^a provided it appears in a cross term and will transform according to the definition $x^a = {x^a}_0 + \alpha^a$ where α^a will appear explicitly as it is not eliminated in any derivation.

If the Lagrangian is independent of the variable x^a its derivative with respect to the variable will be null and therefore the Lagrangian will be constant in that derivative, that is, if $\left(\frac{\partial L}{\partial x^a}\right) = 0$ then $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = 0$ and therefore $\frac{\partial L}{\partial \dot{x}^a} = const.$ that will signify, which is the direct application of Noether's theorem, that we find ourselves in the case of a conserved magnitude, $\frac{\partial L}{\partial \dot{x}^a}$. And therefore we can conclude that a coordinate change in x^a will not alter the conservation of the magnitude associated with that variable, given that $\dot{x}^a = {\dot{x}^a}_0$ and therefore

$$\frac{\partial L}{\partial \dot{x}^a} = \frac{\partial L}{\partial \dot{x}^a_0} = \text{const.}$$

Examples of T3 Derivations

In Minkowski in spherical coordinates with $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + r^2 sin^2\theta d\phi^2$ the term $\frac{\partial L}{\partial x^a}$ will only vary under coordinate changes in r and θ but not in ϕ or t. The conserved magnitudes in t and ϕ will continue to be conserved under coordinate changes in themselves, but not under changes in r and θ.

In Minkowski in Cartesian coordinates $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + dz^2$ the term $\frac{\partial L}{\partial x^a}$ will not vary under coordinate changes in any variable. The conserved magnitudes in the four variables will continue to be conserved under any coordinate change.

In Schwarzschild with $ds^2 = -A(r)c^2dt^2 + B(r)dr^2 + C(r)d\theta^2 + \mathrm{D}(\mathrm{r},\theta)d\phi^2$ the term $\frac{\partial L}{\partial x^a}$ will vary under coordinate changes in r but not in t, θ or ϕ. The conserved magnitudes in t and ϕ will continue to be conserved under coordinate changes in themselves, but not under changes in and θ.

Summary

We are going to integrate the previous terms into the geodesic expression Ec1 taking care not to cause a duplication of indexes that would make them dummy. Thus we have:

(Ec1) $\frac{d}{d\lambda}\left(\frac{\partial \mathrm{L}}{\partial \dot{x}^a}\right) - \frac{\partial L}{\partial x^a} = 0$

Which with T2 = $\frac{d}{d\lambda}\left(\frac{\partial \mathrm{L}}{\partial \dot{x}^a}\right)$ and T3 = $\left(\frac{\partial L}{\partial x^a}\right)$ results in:

$\mathrm{T}_2 - \mathrm{T}_3 = 0$

We checked before that:

T2 = $2\,(\partial_\mu g_{ab})\dot{x}^\mu\,\dot{x}^b\; + 2\, g_{ab}\,\ddot{x}^b$

T3 = $(\partial_a g_{b\mu})\,\dot{x}^b\dot{x}^\mu$

But the indexes in T2 and T3 must be altered to prevent them from violating the tensorial syntax when we subtract T2 and T3 (index b would appear three times in the summation).

We can thus write $2\,(\partial_\mu g_{ab}) = (\,\partial_\mu g_{ab} + \partial_b g_{a\mu})$

To rewrite T2 as T2 = $(\,\partial_\mu g_{ab} + \partial_b g_{a\mu})\dot{x}^\mu\,\dot{x}^b\; + 2\, g_{ab}\,\ddot{x}^b$

Thus Ec1 results in:

$$(\,\partial_\mu g_{ab} + \partial_b g_{a\mu})\dot{x}^\mu\,\dot{x}^b\; + 2\, g_{ab}\,\ddot{x}^b - (\partial_a g_{b\mu})\dot{x}^b\dot{x}^\mu = 0$$

Being the canonical equation of the geodesic according to the principle of least action for the coordinate x^a

(Ec2) $\frac{d^2x^a}{d\lambda^2} + \Gamma^a_{bc}\frac{dx^b}{d\lambda}\frac{dx^c}{d\lambda} = \ddot{x}^a + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$

We want to integrate the previous conclusions into Ec1 to derive how it is altered under the proposed variable change in x^a and how its Christoffel symbols Γ^a_{bc} are altered in particular.

Altering the order of terms in equality Ec1:

$$2\, g_{ab}\,\ddot{x}^b\; + (\,\partial_\mu g_{ab} + \partial_b g_{a\mu})\dot{x}^\mu\,\dot{x}^b - (\partial_a g_{b\mu})\,\dot{x}^b\dot{x}^\mu = 0$$

Factoring out $\dot{x}^b\dot{x}^\mu$ in the second term:

$$2\, g_{ab}\,\ddot{x}^b + (\,\partial_\mu g_{ab} + \partial_b g_{a\mu} - \,\partial_a g_{b\mu})\,\dot{x}^b\dot{x}^\mu = 0$$

And seeking to resemble it to the canonical expression, it is important to take into account that:

$$g^{ad}g_{ab} = \delta^d_b$$

So we are going to multiply the entire expression by a factor (1/2) and the inverse of the metric g^{ad} (we avoid introducing g^{ab} because we want to avoid duplicating the index b and c has already been used for a coordinate in the geodesic), resulting in:

$$\delta^d_b\,\ddot{x}^b \quad + \quad (1/2)g^{ad}(\,\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu})\,\dot{x}^b\dot{x}^\mu = 0$$

And as $\delta^d_b\,\ddot{x}^b$ collapses the index b to d we find that $\delta^d_b\,\ddot{x}^b = \ddot{x}^d$ and the expression results in:

$$\ddot{x}^d \quad + \quad (1/2)g^{ad}(\,\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu})\,\dot{x}^b\dot{x}^\mu = 0$$

And returning to the comparison with the canonical equation:

(Ec2) $\ddot{x}^a + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$

we obtain the analytical expression for the Christoffel symbols:

(Ec3) $\Gamma^d_{b\mu} = (1/2)g^{ad}(\,\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu})$

Thus we see that in equation (Ec2), a rigid change of coordinates in the variable x^a alters the expression of the metric g^{ab} en equation 3 (Ec3) and that of the Christoffel symbols Γ^a and as the geodesic equation is covariant under coordinate changes, the expressions of $\dot{x}^b$ *and* $\dot{x}^c$ must vary with the Christoffel symbol to maintain the equality.

4.6. Canonical Geodesic Equation According to the Principle of Least Action

Here we are going to demonstrate the canonical form of the geodesic equation according to the principle of least action, which is proposed as a good exercise for working with tensorial notation, defining the Lagrangian:

$\mathrm{L} = \frac{ds^2}{d\lambda^2} = \frac{g_{\mu\nu}\,dx^\mu\,dx^\nu}{d\lambda\,d\lambda} = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ with notation $\frac{dx^a}{d\lambda} = \dot{x}^a$ and affine parameter $\frac{dL}{d\lambda} = 0$

Euler-Lagrange least action formulation

$$\frac{d}{d\lambda}\frac{\partial \mathrm{L}}{\partial \dot{x}^\sigma} - \frac{\partial \mathrm{L}}{\partial x^\sigma} = 0$$

Calculation of $\frac{\partial \mathrm{L}}{\partial \dot{x}^\sigma}$

$$\frac{\partial \mathrm{L}}{\partial \dot{x}^\sigma} = \frac{\partial}{\partial \dot{x}^\sigma} g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$$

$$= g_{\mu\nu}\dot{x}^\nu\frac{\partial}{\partial \dot{x}^\sigma}\dot{x}^\mu + g_{\mu\nu}\dot{x}^\mu\frac{\partial}{\partial \dot{x}^\sigma}\dot{x}^\nu$$

and recalling that in tensorial notation $\frac{\partial \dot{x}^a}{\partial \dot{x}^b} = \delta^a_b$

$$\frac{\partial L}{\partial \dot{x}^\sigma} = g_{\mu\nu}\dot{x}^\nu\delta^\mu_\sigma + g_{\mu\nu}\dot{x}^\mu\delta^\nu_\sigma$$
$$= g_{\mu\nu}\left[\delta^\mu_\sigma\dot{x}^\nu + \delta^\nu_\sigma\dot{x}^\mu\right]$$
$$= g_{\sigma\nu}\dot{x}^\nu + g_{\mu\sigma}\dot{x}^\mu = 2g_{\mu\sigma}\dot{x}^\mu$$

Calculation of $\frac{d}{d\lambda}\frac{\partial L}{\partial \dot{x}^\sigma}$

$$\frac{d}{d\lambda}\frac{\partial L}{\partial \dot{x}^\sigma} = \frac{d}{d\lambda}2g_{\sigma\mu}\dot{x}^\mu = 2\frac{d}{d\lambda}g_{\sigma\mu}\dot{x}^\mu$$
$$= 2\dot{x}^\mu\frac{d}{d\lambda}g_{\sigma\mu} + 2g_{\sigma\mu}\ddot{x}^\mu$$

$\frac{d}{d\lambda}g_{\sigma\mu}$ can be expressed as $\partial_a g_{\sigma\mu}\dot{x}^a$, therefore:

$$\frac{d}{d\lambda}\frac{\partial L}{\partial \dot{x}^\sigma} = 2\dot{x}^\mu\,\partial_\nu g_{\sigma\mu}\dot{x}^\nu + 2g_{\sigma\mu}\ddot{x}^\mu$$

Calculation of $\frac{\partial L}{\partial x^\sigma}$

$$\frac{\partial L}{\partial x^\sigma} = \frac{\partial}{\partial x^\sigma}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \dot{x}^\mu\dot{x}^\nu\frac{\partial}{\partial x^\sigma}g_{\mu\nu}$$
$$= \dot{x}^\mu\dot{x}^\nu\,\partial_\sigma g_{\mu\nu}$$

Back to Euler-Lagrange

$$\frac{d}{d\lambda}\frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} = 0 \;\Rightarrow\; \frac{d}{d\lambda}\frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} = 0$$

$$2\dot{x}^\mu\,\partial_\nu g_{\sigma\mu}\dot{x}^\nu + 2g_{\sigma\mu}\ddot{x}^\mu - \dot{x}^\mu\dot{x}^\nu\,\partial_\sigma g_{\mu\nu} = 0$$

Dividing by 2

$$\dot{x}^\mu\,\partial_\nu g_{\sigma\mu}\dot{x}^\nu + g_{\sigma\mu}\ddot{x}^\mu - \frac{1}{2}\dot{x}^\mu\dot{x}^\nu\,\partial_\sigma g_{\mu\nu} = 0$$

By symmetry, $\partial_\nu g_{\sigma\mu}\dot{x}^\nu\dot{x}^\mu$ can be written as $\frac{1}{2}\partial_\mu g_{\nu\sigma}\dot{x}^\mu\dot{x}^\nu + \frac{1}{2}\partial_\nu g_{\mu\sigma}\dot{x}^\mu\dot{x}^\nu$ and therefore

$$\frac{1}{2}\partial_\mu g_{\nu\sigma}\dot{x}^\mu\dot{x}^\nu + \frac{1}{2}\partial_\nu g_{\mu\sigma}\dot{x}^\mu\dot{x}^\nu + g_{\sigma\mu}\ddot{x}^\mu$$
$$-\frac{1}{2}\dot{x}^\mu\dot{x}^\nu\,\partial_\sigma g_{\mu\nu} = 0$$

$$\frac{1}{2}\partial_\mu g_{\nu\sigma}\dot{x}^\mu\dot{x}^\nu + \frac{1}{2}\partial_\nu g_{\mu\sigma}\dot{x}^\mu\dot{x}^\nu - \frac{1}{2}\dot{x}^\mu\dot{x}^\nu\,\partial_\sigma g_{\mu\nu}$$
$$+ g_{\sigma\mu}\ddot{x}^\mu = 0$$

$$\frac{1}{2}\left(\partial_\mu g_{\nu\sigma}\dot{x}^\mu\dot{x}^\nu + \partial_\nu g_{\mu\sigma}\dot{x}^\mu\dot{x}^\nu - \dot{x}^\mu\dot{x}^\nu\,\partial_\sigma g_{\mu\nu}\right)$$
$$+ g_{\sigma\mu}\ddot{x}^\mu = 0$$

$$\frac{1}{2}\dot{x}^\mu\dot{x}^\nu\left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}\right) + g_{\sigma\mu}\ddot{x}^\mu = 0$$

Analytical formulation of Christoffel symbols

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}\left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}\right)$$

Inserting Christoffel symbols into Euler-Lagrange

$$\frac{1}{2}\dot{x}^\mu\dot{x}^\nu\left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}\right) + g_{\sigma\mu}\ddot{x}^\mu = 0$$

$$g_{\sigma\mu}\ddot{x}^\mu + g_{\sigma\lambda}\Gamma^\lambda_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$$

Multiplying by $g^{\rho\sigma}$

$$g^{\rho\sigma}g_{\sigma\mu}\ddot{x}^\mu + g^{\rho\sigma}g_{\sigma\lambda}\Gamma^\lambda_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$$

$$\delta^\rho_\mu\ddot{x}^\mu + \delta^\rho_\lambda\Gamma^\lambda_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$$

$$\ddot{x}^\rho + \Gamma^\rho_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$$

Result in canonical form

$$\ddot{x}^a + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$$

5. Acknowledgements

This document was produced as a final course project for a remote training program provided by the Universidad Nacional de Educación a Distancia (UNED) as part of its continuing education study program, "Geometry of Spacetime. Black Holes and Gravitational Waves" and hence it contains general physics content that exceeds the context of satellite navigation. While the course aimed to explain and apply general relativity in these two cosmological phenomena, I have decided to be humble in my aspirations regarding the syllabus and apply the physics presented to a case where I felt more comfortable and could verify that the conclusions are indeed correct. Thus, I have reserved cosmology to study it without fear of making errors, leaving it for my leisure time.

This course has been my first university-level approach to general relativity and cosmology, and for that, I am extremely grateful to my course professors Carlos Shahbazi Alonso and Pablo Bueno Gómez for their lessons, their initiative in inaugurating and facilitating this course, and all the material they have made available to the students, as well as to my fellow classmates, for being the best and most fortunate company one could hope for in this endeavor.

Should any error be found in the document, it will be the sole responsibility of the author, given that, as can be read, I have gone beyond the required proofs that I could have copied directly from more than one classic text or technical manual, but which I have derived in order to better understand the course contents.

And as a point of interest for those interested readers who are fans of science fiction and probably know the film Interstellar (2014) by Christopher Nolan: at one point in the movie, the protagonists have to separate; some land on a planet near a black hole and others remain far from it, and when they meet again, time has passed much more slowly for those who landed on the planet. They will find a graph that explains it here. I hope you savor this moment as much as I did when I elaborated this graph and thought that its consequences were exactly those shown in the film, which was advised by none other than Kip Thorne, winner of the Nobel Prize in Physics in 2017 for his contribution to the understanding of gravitational waves.

And to point out any of those errors or any other topic of interest relative to it, I leave my contact here: fer.urd@gmail.com.

Fernando Urdiales
Madrid, May 2026